JOINTS

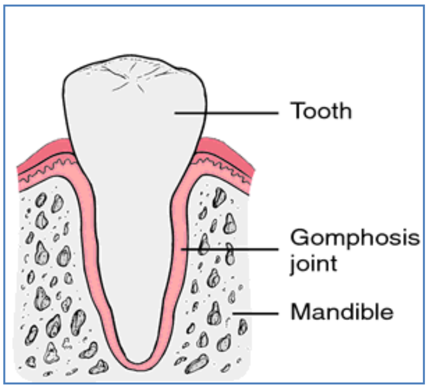
★ **Definition**: The joint is the contact (articulation) between two or more bones together.

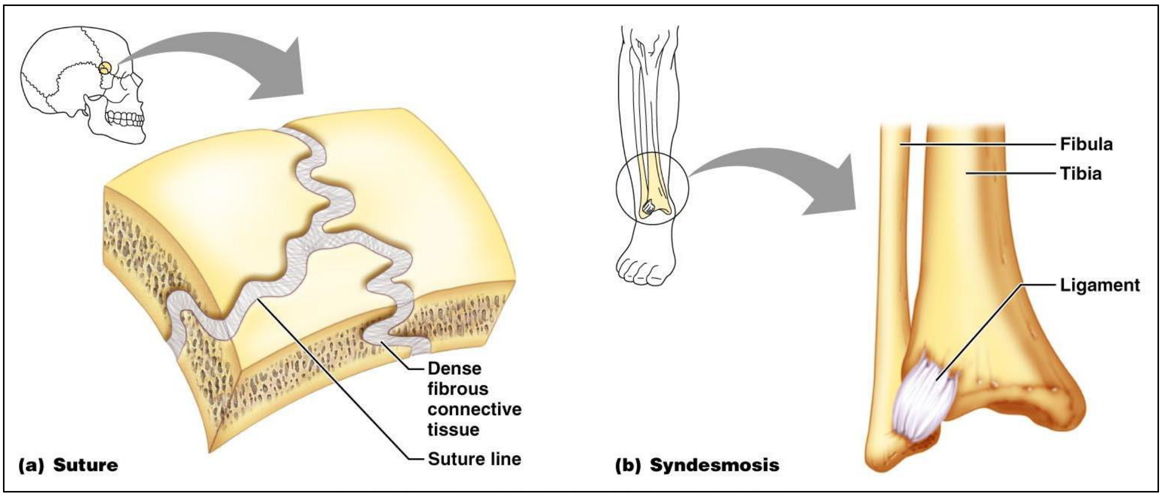
★ **Classification of Joints:** There are 3 types of joints.

I. Fibrous Joints:

- **Fixed** joints in which the surfaces of articulating bones are **connected** together by **fibrous** tissue.
- Fibrous joints are reclassified according to the *amount of the fibrous tissue* into:

1) **Suture:** minimal amount of fibrous tissue.
2) **Gomphosis:** moderate amount
3) **Syndesmosis:** large amount e.g. inferior tibiofibular joint.

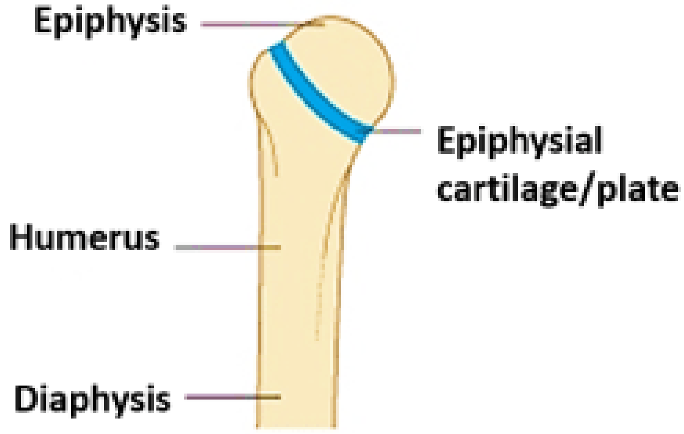




II. The Cartilaginous Joints: (false joint)

- In these joints the **surfaces** of the articulating bones are **connected** together by a disc of **cartilage**.
- They are **2 types**:

1-Primary cartilaginous joint	**2- Secondary cartilaginous joint**
▪ **Temporary** joint disappears by ossification. ▪ **Site:** at the ends of long bones. ▪ **Fixed** joint. ▪ **E.g: Epiphyseal plate.**	▪ **Permanent** joint. ▪ Present in the **midline**. ▪ **N0 or limited** movement which is allowed by the **elasticity** of the fibrocartilaginous **disc**. ▪ E.g: Intervertebral disc.



III. Synovial Joints: (The commonest joints)

- They are **mobile** joints present mostly in the **limbs.**
- They allow fee range of movements.
- **Structure of synovial joints:**

1) Fibrous capsule

- The synovial joint is **surrounded** completely by a fibrous capsule which is **lined by** synovial membrane.
- This capsule is **supported** and strengthened by **ligaments.**

2) Articular cartilage:

- **Hyaline** cartilage **covers** the articular surfaces of bones.
- It is very **smooth** and is **lubricated** by the synovial fluid.
- **Nutrition**: From synovial fluid.
- It has **no** blood or nerve supply.
- **Applied anatomy:** In old age it shows irregularities and erosions. The eroded areas do not repair with pain & limitation of movements. This disease is called osteoarthritis .

3) Joint cavity:

- It is a **potential** cavity which appears if fluid, blood or pus collects into it.
- Normally it **contains** a very thin film of synovial fluid.

4) Synovial membrane :

- Thin, moist, **smooth** and glistening membrane that **covers** all structures inside the joint **except** the articular surfaces.
- It also **lines** the fibrous capsule.
- It secretes and absorbs the **synovial fluid**.

5) **Synovial fluid:**

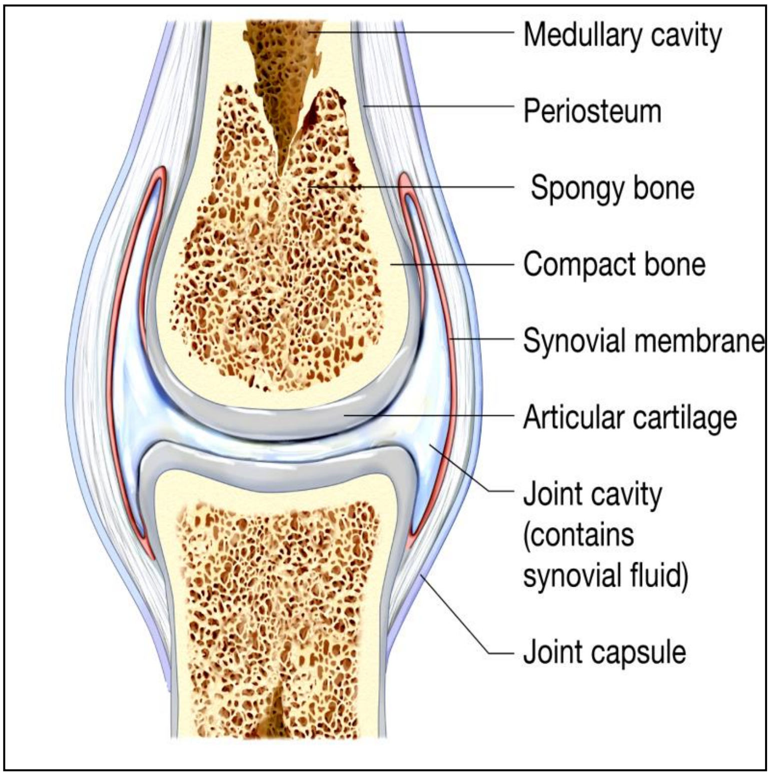
- Pale yellow **viscous fluid** similar to egg-albumin.
- **Functions:**
  - **Lubrication & nutrition of** the articular cartilage.
  - Prevents **erosion** of articular cartilage.

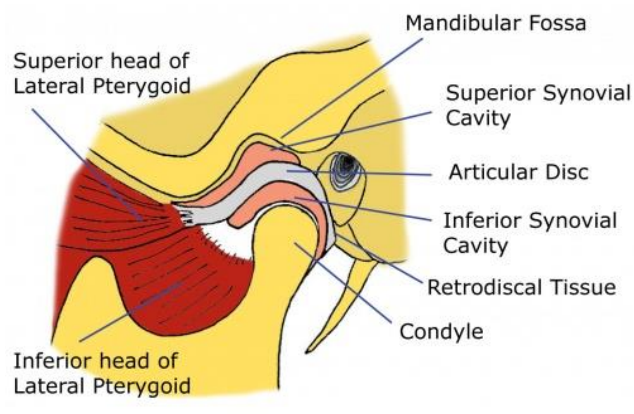
6) **Ligaments:**

- **Extracapsular** and **intracapsular** ligaments which **support** and **strength** the joint.

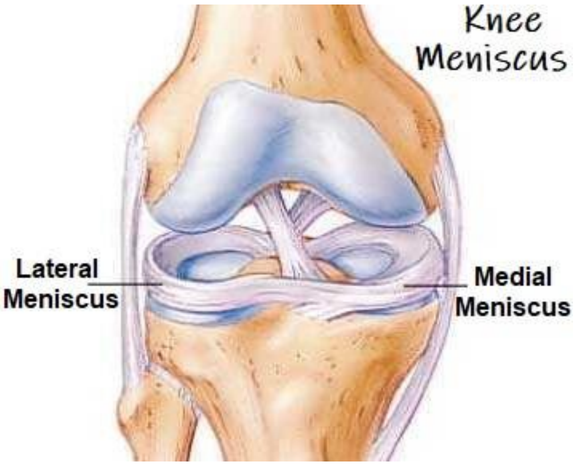
7) **Structures** which may be present **inside the cavity of synovial joints:**

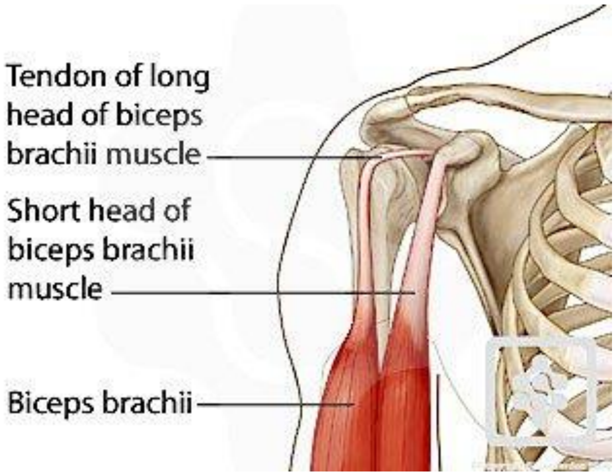
1- **Articular disc:** Disc of **fibrocartilage** which **divides** the joint cavity into two compartments (upper and lower) e.g. temporo-mandibular joint.





The Temporomandibular Joint





2- **Menisci (semilunar cartilages):** e.g. the two semilunar plates of **fibrocartilage** present **inside** the knee joint.

3-**Ligament:** e.g. **cruciate** ligament inside the **knee** joint.

4-**Tendon:** e. g. the tendon of long head of **biceps** muscle inside the **shoulder** joint.

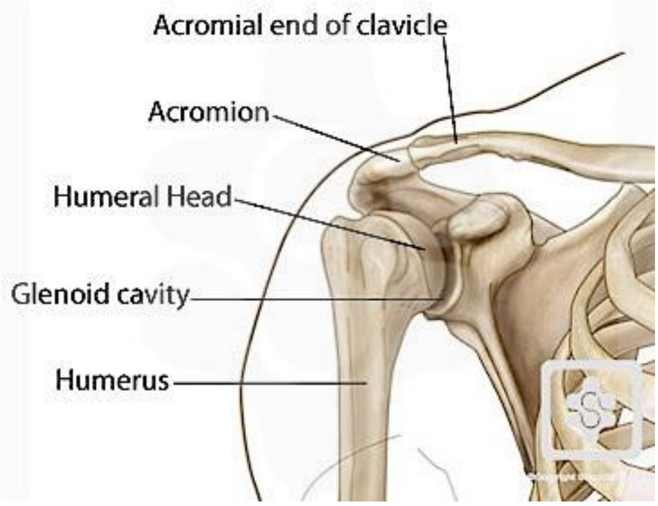
★ **Factors affecting the stability of the joint:**

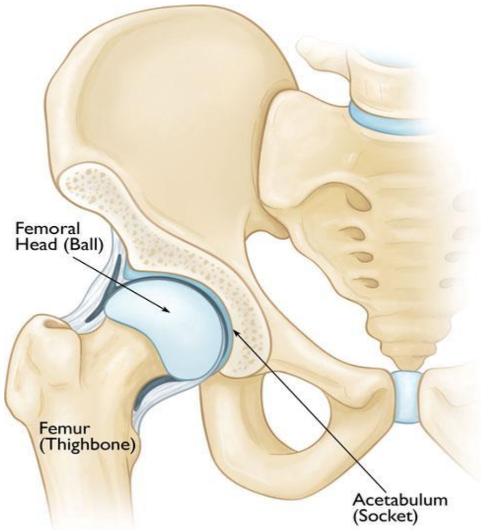
1- Shape and fitting of **articulating surfaces**.

2- Thickness and strength of the **capsule**.

3- Position and strength of **ligaments**.

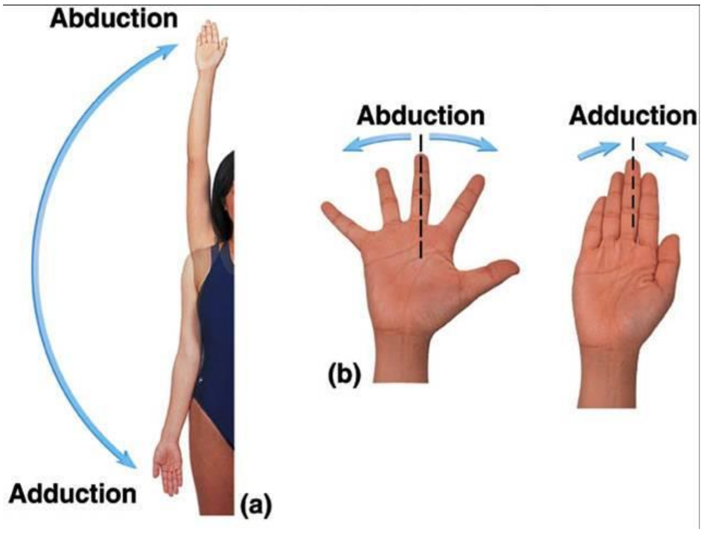
4- Strength of **muscles** surrounding the joint**.**



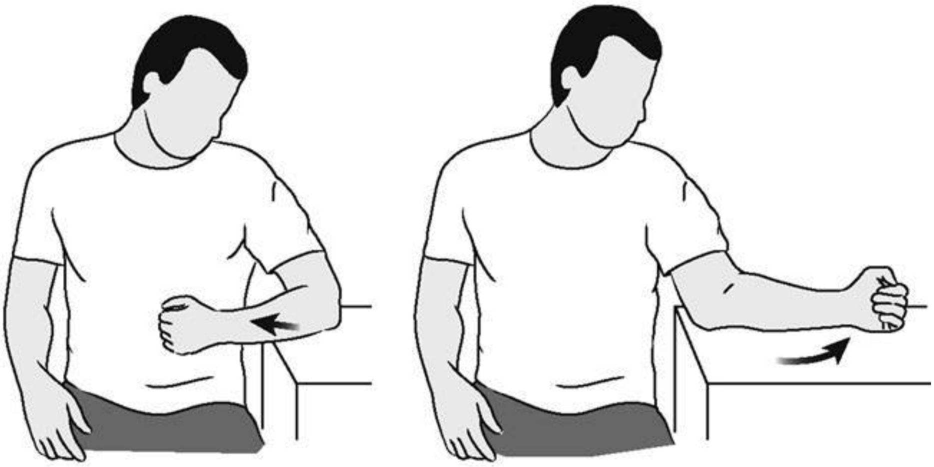


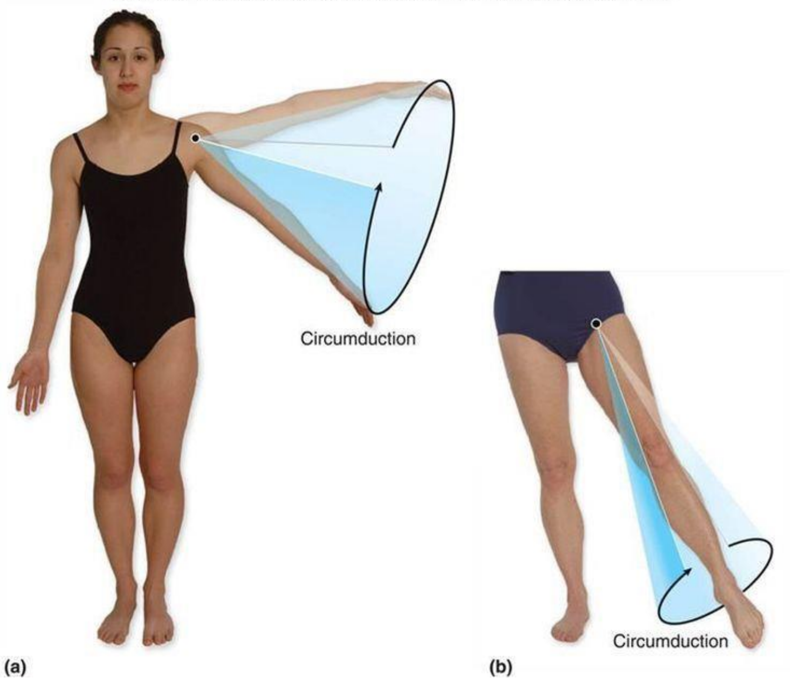
Movements of the Joints

1. **Flexion:** Approximation of two anterior surfaces to each others (bending).
2. **Extension:** Approximation of two posterior surfaces to each others (Straightening).
3. **Abduction:** Movement of the limb laterally away from the middle line.
4. **Adduction:** Movement of the limb medially towards the middle line.



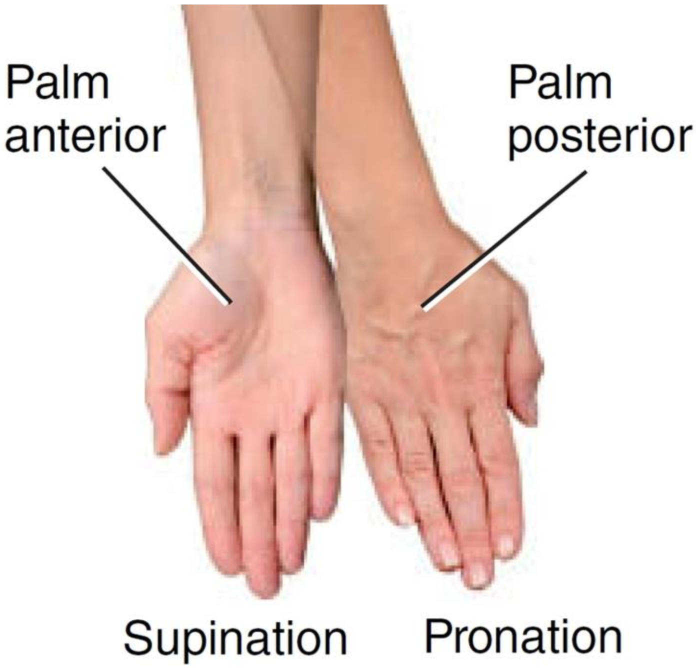
5. **Rotation:** Medial or lateral rotation of the limb around a vertical axis.
6. **Circumduction:** Combination of all above movements.





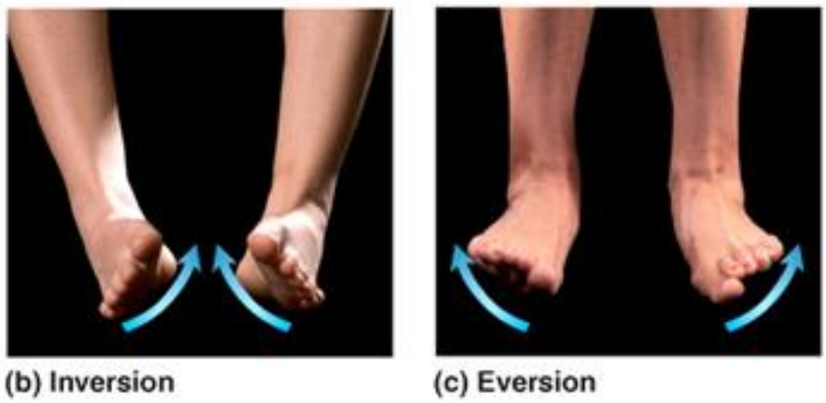
7. **Supination:** The lateral rotation of the forearm.

8. **Pronation:** The medial rotation of the forearm.

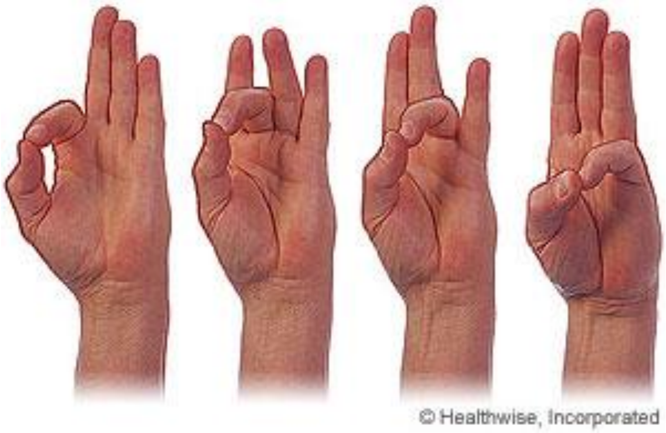


9. **Inversion:** The sole of foot is directed medially.

10. **Eversion:** The sole of foot is directed laterally.



11. **Opposition:** The thumb come in contact with the other 4 fingers.



★ **Classification of Synovial Joints:**

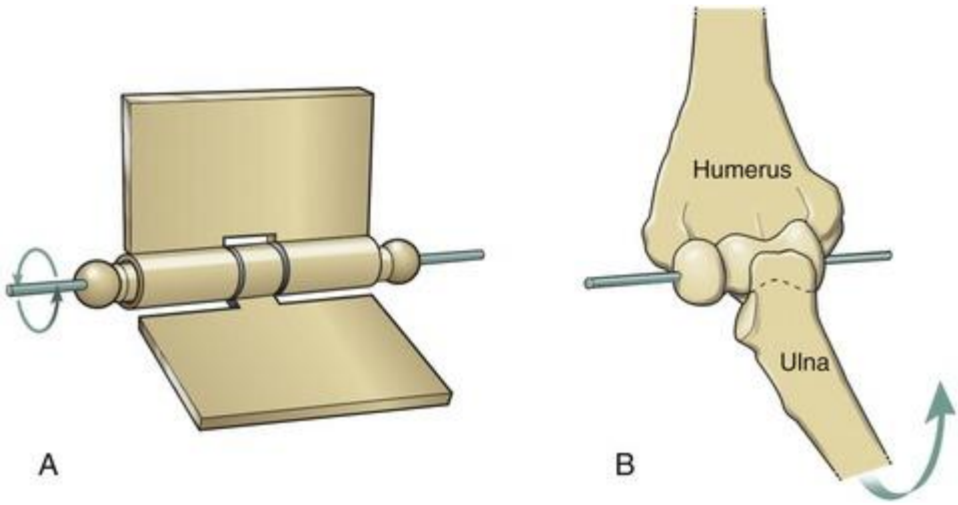
A- According to the axis of movement:

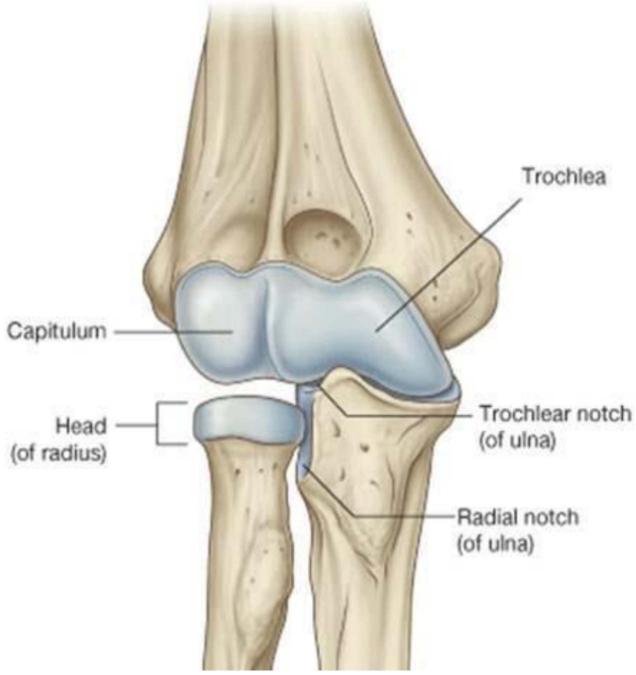
I. Uni-axial joints:

- In this type of joints the movements take place around a **single axis**.
- According to the **direction of axis**: there are 2 types:

1-Hinge joint:

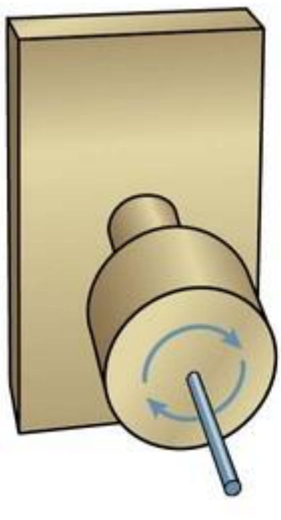
- The joint is uni-axial with **one transverse axis** e.g. elbow joint .
- The **movement** is flexion and extension only with no abduction or adduction e.g. elbow.



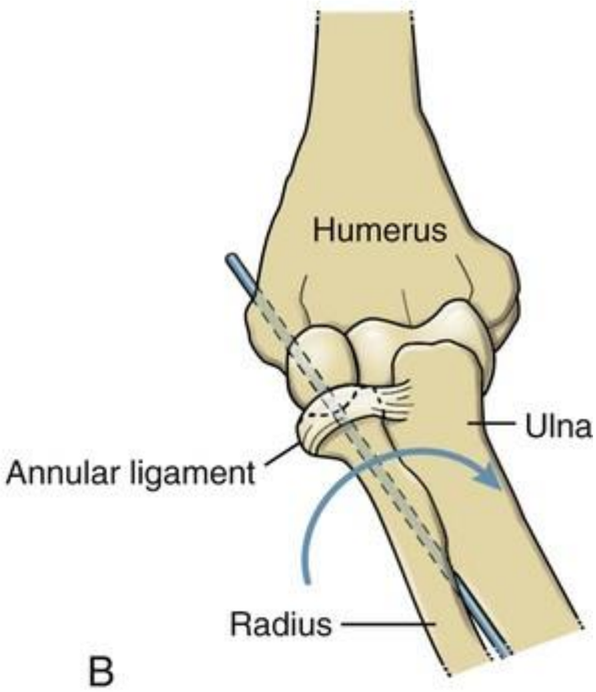


2-Pivot joint:

- The joint is uni-axial with **one** vertical **axis** e.g. superior radio-ulnar joint.
- The **movement** of this joint is rotation around the center of the pivot. i.e **pronation and supination** in superior and inferior radio-ulnar joint.



A



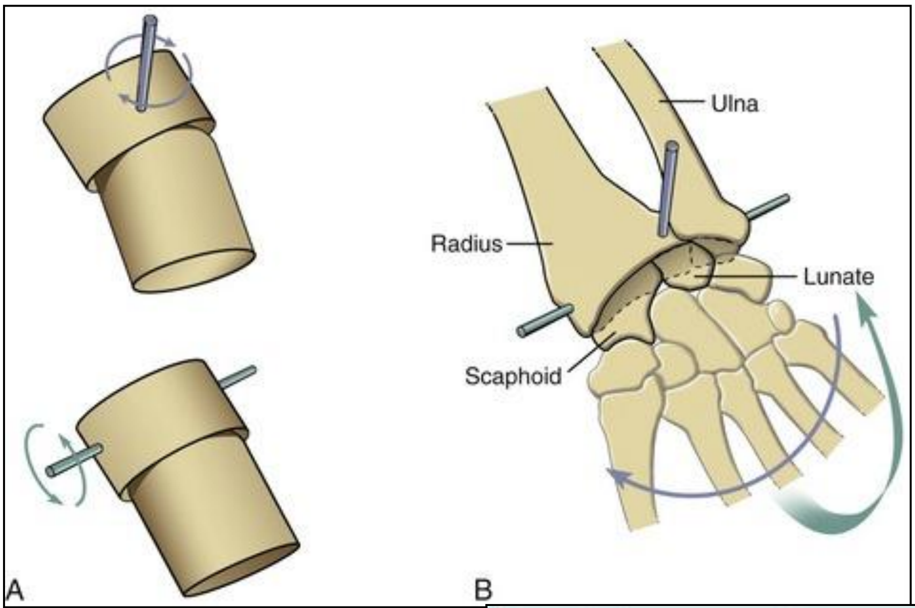
B

II. Bi-axial joints:

- In this type of joints the movements take place around **2 axes perpendicular to each other.**
- According to the **shape of the articular surface** there are **3 types:**

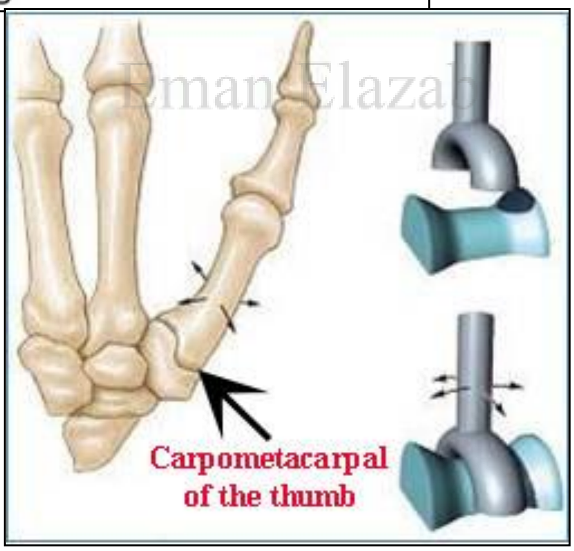
1- **Ellipsoid joint**

- It is formed by the articulation of an **oval convex** surface (carpal bones) with an **elliptical concave** surface (inferior surface of radius and the articular disc of ulna) e.g. **wrist joint**.
- The **movements** of this joint are flexion, extension, abduction, adduction and circumduction.



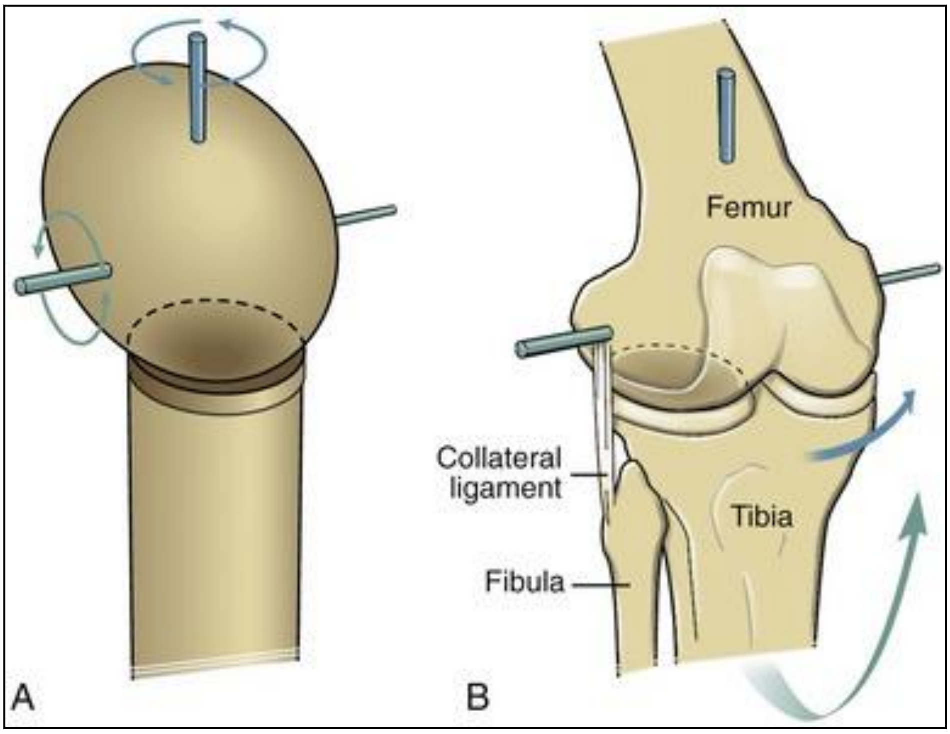
2- **Saddle joints**

Each articular surface of each bone is convex in one direction and concave in the perpendicular direction.



3- **Modified hinge**

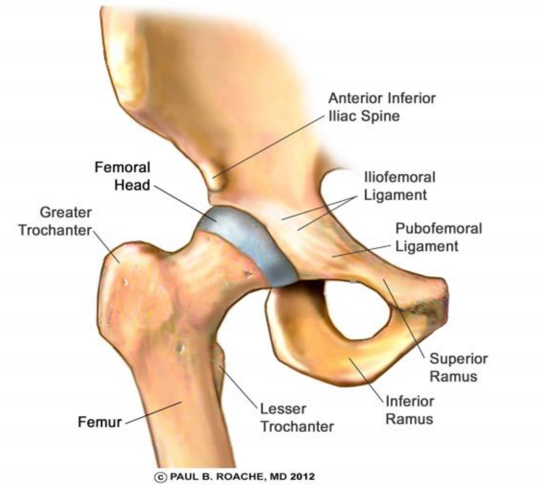
- It consists of either **2** separate convex surfaces (condyles) which articulate with 2 concave surfaces e.g. ***knee joint*** or **one** condyle articulating with one concave surface e.g. ***temporo-mandibular joint***.
- The **movements** of this joint are flexion, extension and rotation (medial and lateral)

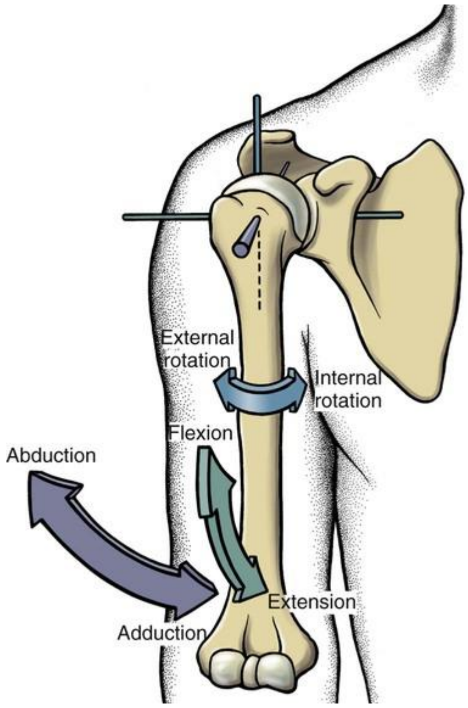


III. Multi-axial (polyaxial) joints:

- In this type of joints, the movements occur around **3 axes.**
- The **articuclar surface** consists of rounded bone articulates with **concave socket**. According to the **shape** of the articular surfaces, they are called **ball and socket joints** e.g. **shoulder** and **hip** joints.
- They are the **most freely mobile** joints in the body.
- The **movements** of this joint are around **transverse axis (**flexion & extension), around **antero-posterior axis** (abduction

& adduction), around **vertical axis** (medial & lateral rotation) and circumduction.

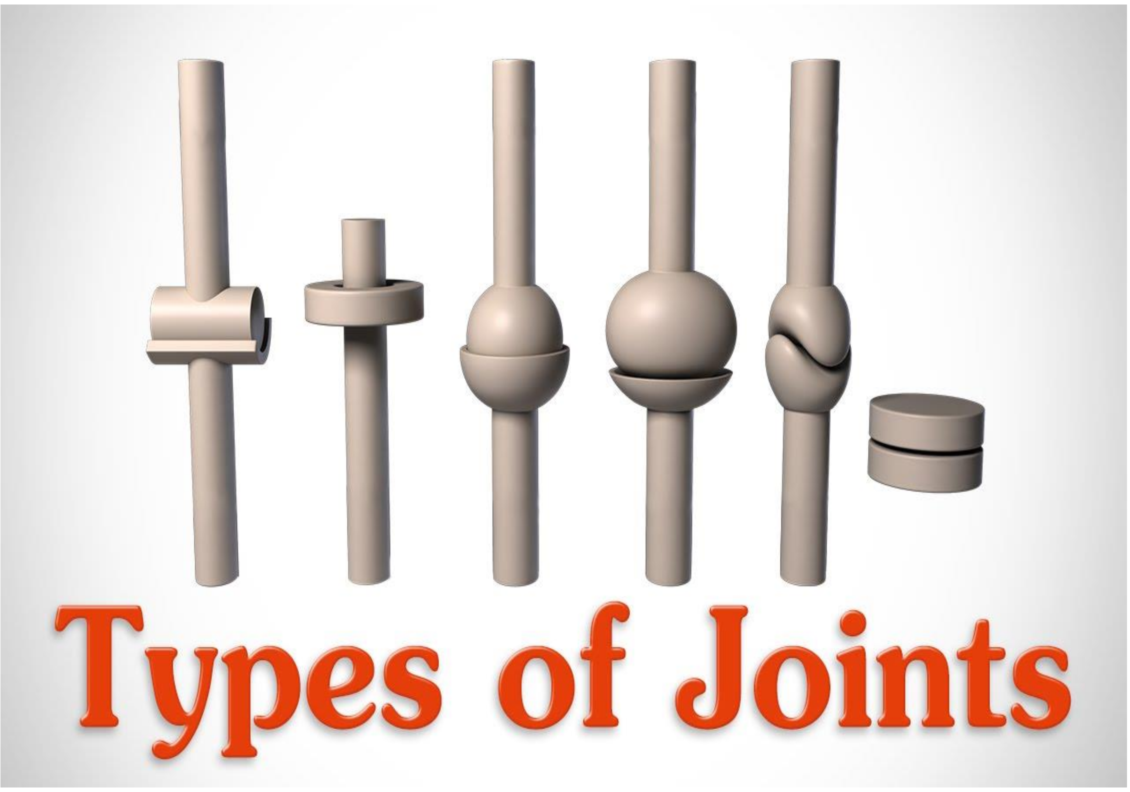




★ Please see this excellent video :
https://www.youtube.com/watch?v=0cYal_hitz4

Summary of types of synovial joints

According to the axis of movement.	**According to shape of the articular surface**
I. Uni-axial joints	1- Hinge e.g. elbow joint
	2- Pivot e.g. superior R.U. joint
II. Bi-axial joints	1- Ellipsoid e.g. wrist joint
	2- Saddle e.g. carpometacarpal of thumb
	3- Modified hinge e.g. knee joint
III. Multi-axial joints	★ Ball and socket e.g. Hip & shoulder

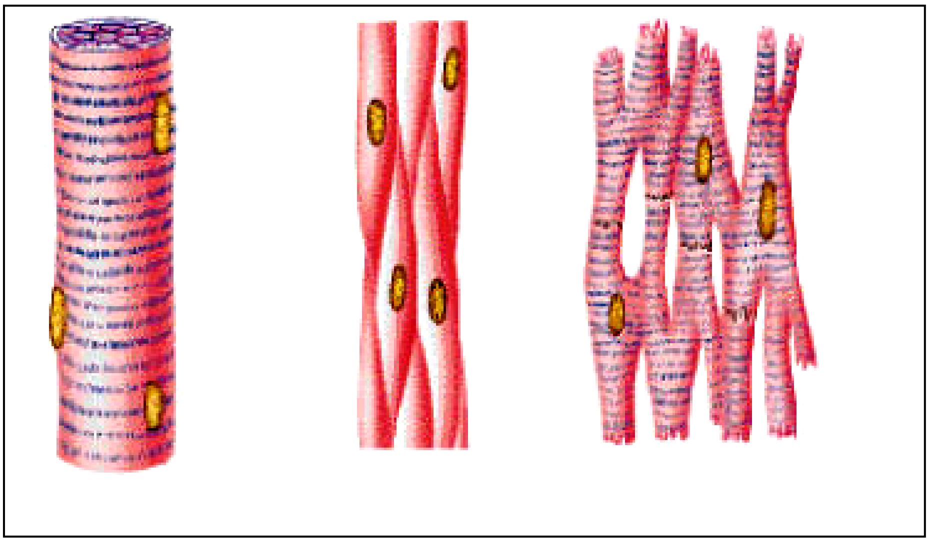

Types of Joints

Muscular System

⁕ Muscle tissue is characterized by the property of **contraction** which is the ability of the muscle fibers to **become short**.

⁕ **Classification of the muscles** according to the structure and function, there are 3 types of:

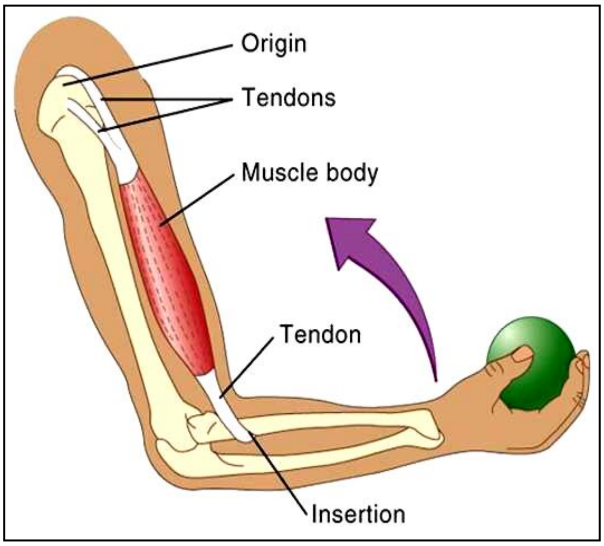
Muscle	I. Skeletal	II. Smooth	III. Cardiac
1- Site	Attached to skeleton (bones)	In the wall of blood vessels and viscera.	In the myocardium.
2- Contraction	Voluntary	Involuntary	Involuntary
4- Nerve supply	Somatic nerve	Autonomic nerve	Autonomic nerve



Skeletal. Smooth. Cardiac M.

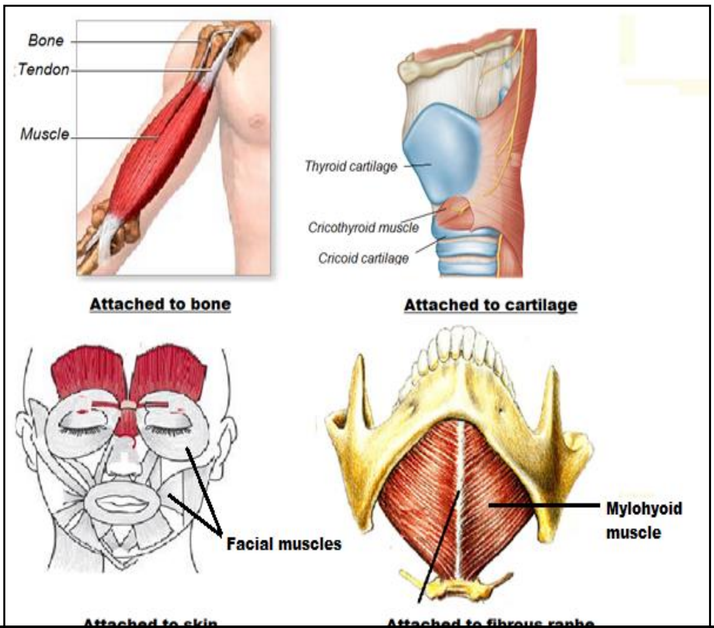
⁕ **Attachment of skeletal muscles**: Each muscle has **two** attachments:

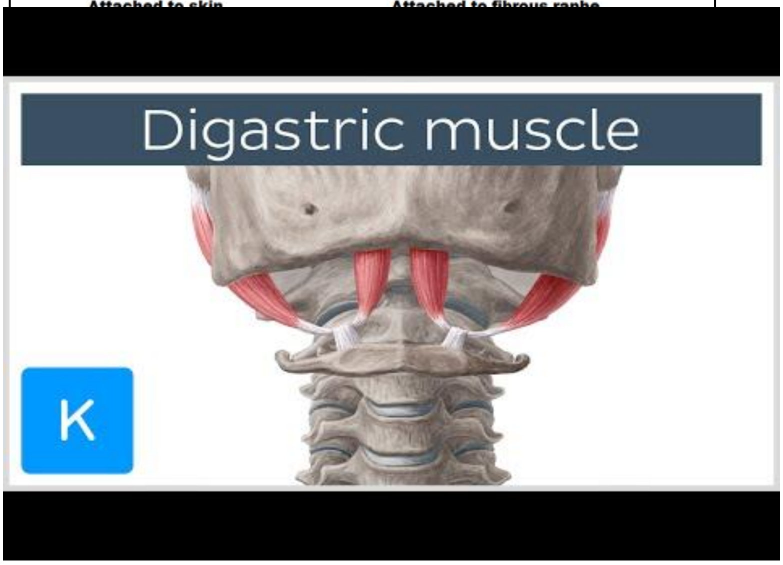
- **Origin:** the most fixed attachment.
- **Insertion:** The most mobile attachment.



⁕ **Types of Muscle Attachment:**

1. **Attachment to bone:** It is the **commonest** type. A muscle may be attached to bone either by fleshy fibers or by a tendon.
2. **Attachment to a fibrous raphe:** It is a **linear band of fibrous tissue** through which 2 muscles fuse together e.g. mylohyoid muscle.
3. **Attachment to skin:** A muscle is inserted into the dermis of the skin and by its contraction it could move the skin, e.g. facial muscles.
4. **Attachment to cartilage:** as in the muscles of the larynx e.g. cricothyroid muscle.
5. **Intermediate tendon** joins 2 fleshy bellies together e.g. digastric muscle.

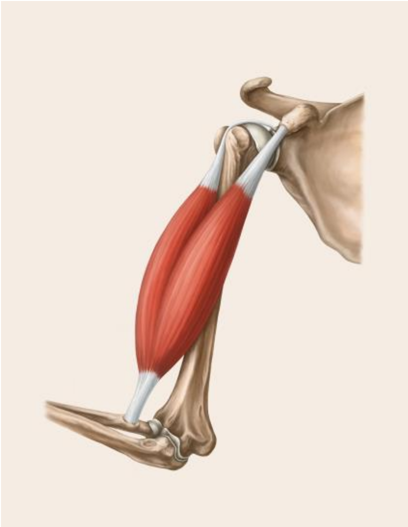


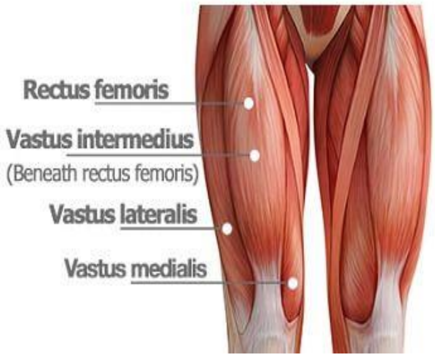


Types of muscle attachment

⁕ Muscles having **more than one head**:

1. A muscle having **two** heads e.g. **bi**ceps muscle.
2. A muscle having **three** heads e.g. **tri**ceps muscle.
3. A muscle having **four** heads e.g. **quadr**iceps muscle.





Biceps muscle **Triceps muscle** **Quadriceps muscle**

Muscles having more than one head

⁕ **Clinical Importance:** Muscle atrophy occurs in:

1- Disuse atrophy.

2- Immobilization after fracture.

3- Disease of the muscle itself.

4- Injury of the motor nerve or arterial supply of the muscle.

5- Injury of spinal cord.

6- Affection of higher centers (brain) e.g. hemiplegia,….etc.